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|-----------------------|-----------------------|---------------------------------|----------------------------------|
| <b>13</b><br>Chapters | <b>245</b><br>Periods | <b>204h 10min</b><br>Total time | <b>245×50min</b><br>Period types |
|-----------------------|-----------------------|---------------------------------|----------------------------------|

## Executive summary

This report presents the allocation of available instructional periods across the selected chapters for Science, Grade IX. Aruvi looks at the teaching effort each chapter is likely to need — how abstract each chapter's ideas are, how much sustained reasoning it demands, and the in-class workload of its exercises — and turns that into a single effort index for the chapter. Chapters with a higher effort index are given a larger share of the available periods, so that harder chapters get more classroom time, and periods are then distributed across all the selected chapters in proportion to their effort index. The allocation report below shows the distribution of the available periods across the chapters using the approach described here.

For each chapter, the competency section lists the competencies the chapter addresses, together with the rationale for including each one.

This information is intended to help the teacher in planning the class time, aligning with curriculum guidelines, and to provide an audit trail for the same.

## Allocation details

| #            | Chapter                                              | 50-min Periods | Total Periods |
|--------------|------------------------------------------------------|----------------|---------------|
| 1            | Exploration: Entering the World of Secondary Science | 6              | <b>6</b>      |
| 2            | Cell: The Building Block of Life                     | 15             | <b>15</b>     |
| 3            | Tissues in Action                                    | 15             | <b>15</b>     |
| 4            | Describing Motion Around Us                          | 22             | <b>22</b>     |
| 5            | Exploring Mixtures and their Separation              | 18             | <b>18</b>     |
| 6            | How Forces Affect Motion                             | 21             | <b>21</b>     |
| 7            | Work, Energy, and Simple Machines                    | 22             | <b>22</b>     |
| 8            | Journey Inside the Atom                              | 12             | <b>12</b>     |
| 9            | Atomic Foundations of Matter                         | 22             | <b>22</b>     |
| 10           | Sound Waves: Characteristics and Applications        | 18             | <b>18</b>     |
| 11           | Reproduction: How Life Continues                     | 15             | <b>15</b>     |
| 12           | Patterns in Life: Diversity and Classification       | 15             | <b>15</b>     |
| 13           | Earth as a System: Energy, Matter, and Life          | 9              | <b>9</b>      |
| <b>Total</b> |                                                      | <b>210</b>     | <b>210</b>    |

# Competency report

## Ch 01 Exploration: Entering the World of Secondary Science

6×50min · 6 periods · 300min

| # | Code         | Competency                                                                                                                                                                                                                                                                                                                                                                              | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-8.1</b> | Develops accurate and appropriate models (including geometric, mathematical, graphical) to represent real-life events and phenomena using scientific principles and use these models to manipulate variables and predict results                                                                                                                                                        | The chapter's central instructional strand develops the concept of scientific models as deliberate simplifications that retain only the details relevant to the question at hand. It introduces the assumptions behind model-building (Example 1.1, Activity 1.1), the use of mathematical relationships to express and test models, and the role of predictions as reasoned expectations derived from laws and theories — directly mapping to C-8.1's requirement to develop models to represent real-life events and use them to predict results.                |
| 2 | <b>C-8.2</b> | Designs and implements a plan for scientific inquiry (formulates hypotheses, makes predictions, identifies variables, accurately uses scientific instruments, represents data — primary and secondary — in multiple modes, draws inferences based on data and understanding of scientific concepts, theories, laws, and principles, communicates findings using scientific terminology) | A distinct strand of the chapter addresses the design and evaluation of scientific inquiry: what makes a prediction scientifically testable (Example 1.2), the role of measurable evidence and past patterns in forming hypotheses, the skill of estimation to check whether answers are reasonable (Example 1.3), and the distinction between laws, theories, and principles as products of systematic inquiry. These map directly to C-8.2's formulation of hypotheses, identification of variables, and communication of findings using scientific terminology. |

## Ch 02 Cell: The Building Block of Life

15×50min · 15 periods · 750min

| # | Code         | Competency                                                                                                                                                                                                                                                                | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-3.1</b> | Explains the role of cellular components (nucleus, mitochondria, endoplasmic reticulum, vacuoles, chloroplast, cell wall), including the semipermeability of cell membrane in making cell the structural basis of living organisms and functional basis of life processes | The chapter's bulk explains the role of cellular components — nucleus, mitochondria, ER, Golgi apparatus, ribosomes, lysosomes, plastids, vacuoles, and cell wall — and specifically the semipermeability of the cell membrane (osmosis, fluid-mosaic model, isotonic/hypotonic/hypertonic effects) in making the cell the structural and functional unit of life. Activities 2.2 and 2.3 directly operationalise selective permeability. Sections 2.2–2.3.1 and 2.5 are the chapter's dominant instructional strand.                                                    |
| 2 | <b>C-3.3</b> | Describes mechanisms of heredity (in terms of DNA, genes, chromosomes) and variation (as changes in the sequence of DNA)                                                                                                                                                  | A distinct, sectioned strand (Nucleus — House of coded instructions; Section 2.4 How do Normal Cells Grow and Divide?; Section 2.4.1 Cell division; Mitosis; Meiosis) establishes DNA, genes, and chromosomes as the carriers of heredity, explains how mitosis preserves genetic identity across body cells, and how meiosis halves chromosome number to produce gametes — with fertilisation restoring the full count and introducing variation. This strand is cognitively independent from C-3.1 and constitutes a distinct, co-central competency in its own right. |

## Ch 03 Tissues in Action

15×50min · 15 periods · 750min

| # | Code         | Competency                                                                                                                                                                                                                                                                | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-3.1</b> | Explains the role of cellular components (nucleus, mitochondria, endoplasmic reticulum, vacuoles, chloroplast, cell wall), including the semipermeability of cell membrane in making cell the structural basis of living organisms and functional basis of life processes | The chapter's dominant instructional strand is the structure–function relationship of tissue types at the cellular level: meristematic cell characteristics (thin walls, prominent nucleus, dense cytoplasm, absence of vacuoles) that enable continuous division; the cellular basis of differentiation into permanent tissues; and the detailed structural descriptions of every tissue type — epidermis, parenchyma, collenchyma, sclerenchyma, xylem/phloem components, epithelial variants, connective tissue matrix differences, and the three muscle types (skeletal, smooth, cardiac) plus the neuron — all showing how cell-level features (wall composition, shape, branching, striations, multinucleation) make each tissue the structural and functional basis of its specific life role. This is the C-3.1 domain of cellular components making cells the structural and functional basis of living organisms. |

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|---|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | <b>C-3.2</b> | Analyses similarities and differences in the life processes involved in nutrition (photosynthesis in plants; absorption of nutrients in fungi; digestion in animals), transport (transport of water in plants; circulation in animals), exchange of materials (respiration and excretion), and reproduction | A distinct, sectioned strand running through section (iii) Conducting tissues and section 3.3.1 directly addresses life processes named in C-3.2: xylem transporting water and minerals from roots upward (transport of water in plants), phloem transporting assimilates from leaves to other parts (food transport), and epithelial tissue enabling rapid gas and liquid diffusion at lung and blood-vessel linings (exchange of materials) and efficient nutrient/water absorption at the small intestine lining (absorption). These are covered in their own named subsections and constitute a cognitively independent strand from the structural catalogue of C-3.1. |
|---|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Ch 04 Describing Motion Around Us

22×50min · 22 periods · 1100min

| # | Code         | Competency                                                                                                                                                                                                                                                                         | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-2.1</b> | Applies Newton's laws to explain the effect of forces (change in state of motion — displacement and direction, velocity and acceleration, uniform circular motion, acceleration due to gravity) and analyses graphical and mathematical representations of motion in one dimension | The chapter's entire domain is the description of motion in one dimension and uniform circular motion — exactly the subject matter named in C-2.1. Sections 4.1–4.1.4 develop displacement and direction, average speed, average velocity, and average acceleration; Section 4.2 builds graphical representations (position-time and velocity-time graphs) and teaches how to extract velocity and acceleration from slopes and displacement from areas; Section 4.3 derives and applies kinematic equations, which are the mathematical representations of motion; Section 4.4.1 develops uniform circular motion including the role of continuously changing velocity direction. Acceleration due to gravity appears explicitly in Example 4.4. No second competency strand exists: the chapter never enters force mechanics (Newton's laws proper), optics, circuits, energy, or sound. |

## Ch 05 Exploring Mixtures and their Separation

18×50min · 18 periods · 900min

| # | Code         | Competency                                                                                                                                                                                            | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-1.2</b> | Investigates the nature and properties of chemical substances (distillation, crystallisation, chromatography, centrifugation, types and properties of mixtures, solutions, colloids, and suspensions) | The chapter's entire domain is the investigation of chemical substances — their classification into solutions, suspensions, and colloids; the quantitative expression of concentration (% m/m, % m/v, % v/v); solubility and its temperature dependence; and the selection and application of separation techniques including crystallization, distillation, paper chromatography, separating funnel (immiscible liquids), sublimation, centrifugation, and coagulation. Every topic is a facet of C-1.2's explicit mandate — 'types and properties of mixtures, solutions, colloids, and suspensions' and the separation techniques named therein. No distinct strand belonging to C-1.1 (atomic structure/valency) or C-1.3 (chemical equations) is present. |

## Ch 06 How Forces Affect Motion

21×50min · 21 periods · 1050min

| # | Code         | Competency                                                                                                                                                                                                                                                                         | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-2.1</b> | Applies Newton's laws to explain the effect of forces (change in state of motion — displacement and direction, velocity and acceleration, uniform circular motion, acceleration due to gravity) and analyses graphical and mathematical representations of motion in one dimension | The chapter's entire domain is Newton's three laws of motion and their application to explaining changes in state of motion — the core clause of C-2.1. Sections 6.1–6.3 build the concepts of force, balanced/unbalanced forces, and friction as the physical cause of deceleration. Sections 6.4–6.6 state and apply Newton's first, second, and third laws, including graphical analysis of position-time and velocity-time graphs and the mathematical relation $F = ma$ . Section 6.7 extends Newton's second law to a system of two connected objects. No distinct second strand exists: the $F = mg$ derivation is instrumental to Newton's second law, not an independent treatment of gravitation (C-2.2), and simple machines (C-2.6) and work-energy (C-2.5) are absent. |

## Ch 07 Work, Energy, and Simple Machines

22×50min · 22 periods · 1100min

| # | Code | Competency | Justification |
|---|------|------------|---------------|
|---|------|------------|---------------|

|   |              |                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-2.5</b> | Defines work in scientific terms, and represents the relationship between potential and kinetic energy (conservation of energy) in mathematical expressions | The chapter's first and larger strand (sections 7.1–7.4.3) defines work in scientific terms ( $W = F \times s$ ), derives kinetic energy ( $K = \frac{1}{2}mv^2$ ) and gravitational potential energy ( $U = mgh$ ) using the work-energy theorem, and mathematizes the conservation of mechanical energy. This directly addresses the 'defines work in scientific terms, and represents the relationship between potential and kinetic energy (conservation of energy) in mathematical expressions' clause of C-2.5.                                                        |
| 2 | <b>C-2.6</b> | Demonstrates the principle of mechanical advantage by constructing simple machines (system of levers and pulleys)                                           | Section 7.6 constitutes a distinct second strand that develops mechanical advantage through hands-on construction and analysis of three simple machines — pulley (fixed and movable), inclined plane, and lever. Activities 7.3 (inclined plane spring-balance experiment), 7.4 (lever with ruler and coins), and 7.5 (beam balance) directly address the 'demonstrates the principle of mechanical advantage by constructing simple machines (system of levers and pulleys)' clause of C-2.6. This strand is cognitively and structurally independent of the energy strand. |

## Ch 08 Journey Inside the Atom

12×50min · 12 periods · 600min

| # | Code         | Competency                                                                                                                                                                                       | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-1.1</b> | Describes classification of elements in the Periodic Table, and explains how compounds (including carbon compounds) are formed based on atomic structure (Bohr's model) and properties (valency) | The chapter's whole domain is atomic structure via the Bohr model and the combining capacity (valency) that follows from electronic configuration — the 'atomic structure (Bohr's model) and properties (valency)' clause of C-1.1. The periodic-table / compound-formation clause of C-1.1 is not developed here; it is the domain of the sibling chapter (Atomic Foundation of Matter). The chapter's four sub-topics (model evolution, atomic/mass number and electronic configuration, valency, isotopes/isobars) are all facets of atomic structure and serve this single competency; no distinct second strand exists within CG-1 (mixtures C-1.2 and chemical equations C-1.3 are absent). |

## Ch 09 Atomic Foundations of Matter

22×50min · 22 periods · 1100min

| # | Code         | Competency                                                                                                                                                                                       | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-1.1</b> | Describes classification of elements in the Periodic Table, and explains how compounds (including carbon compounds) are formed based on atomic structure (Bohr's model) and properties (valency) | The chapter's whole domain is how atoms combine to form compounds — the central clause of C-1.1. The two empirical laws (conservation of mass, constant proportions) and Dalton's atomic theory serve as the foundational scaffolding that explains why atoms combine in fixed ratios. The bulk of the chapter then develops exactly the two bonding mechanisms the C-1.1 clause refers to: covalent bonding (electron sharing in $H_2$ , $Cl_2$ , $O_2$ , $HCl$ , $H_2O$ ) and ionic bonding (electron transfer in $NaCl$ , $MgCl_2$ , etc.), together with the valency-based criss-cross method for writing chemical formulae, naming conventions, properties of the resulting compounds, and the mass calculations (molecular mass and formula unit mass) that follow from knowing the formula. No distinct second strand within CG-1 is present: C-1.2 (mixtures and separation) and C-1.3 (chemical equations with acids/bases/metals/non-metals) are both absent as taught content. The Laws of Conservation of Mass and Constant Proportions appear as historical motivation and verification exercises, not as a developed C-1.3 strand of balanced equations and reaction-type classification. |

## Ch 10 Sound Waves: Characteristics and Applications

18×50min · 18 periods · 900min

| # | Code | Competency | Justification |
|---|------|------------|---------------|
|---|------|------------|---------------|

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|---|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-2.7</b> | Describes the origin and properties of sound (wavelength, frequency, amplitude) and differences in what we hear as it propagates through different instruments | The chapter's entire domain is the origin, propagation, and quantitative characteristics of sound waves — directly the 'describes the origin and properties of sound (wavelength, frequency, amplitude)' clause of C-2.7. All major sections (production via vibration, compression-rarefaction wave mechanics, graphical representation, wave equation $v = \lambda\nu$ , pitch/loudness, reflection leading to echo and reverberation, and ultrasound/infrasound applications including echolocation and sonar) are facets of this single competency. No second C-code within CG-2 is substantively addressed: C-2.5 (energy conservation, work) is referenced only at the level of 'sound carries energy' without mathematical treatment of potential/kinetic energy, and no other CG-2 domain strand is developed as an independent section. |
|---|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Ch 11 Reproduction: How Life Continues

15×50min · 15 periods · 750min

| # | Code         | Competency                                                                                                                                                                                                                                                                                                  | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-3.2</b> | Analyses similarities and differences in the life processes involved in nutrition (photosynthesis in plants; absorption of nutrients in fungi; digestion in animals), transport (transport of water in plants; circulation in animals), exchange of materials (respiration and excretion), and reproduction | The chapter's entire domain is reproduction as a life process — it systematically traces asexual reproduction (vegetative propagation, budding, spore formation) and sexual reproduction across the spectrum from plants (flower structure, pollination, fertilisation, seed formation) through animals (external vs. internal fertilisation, variation in reproductive strategies) to humans (male and female reproductive systems, gametogenesis, ovulation, pregnancy, childbirth, contraception). This matches C-3.2's 'reproduction' clause directly. All six student activities (11.1–11.6) operationalise aspects of this reproductive life process from propagation techniques to pollination mechanics.                     |
| 2 | <b>C-3.3</b> | Describes mechanisms of heredity (in terms of DNA, genes, chromosomes) and variation (as changes in the sequence of DNA)                                                                                                                                                                                    | Section 11.2.1 constitutes a cognitively distinct, named sectioned strand on the mechanisms of heredity and variation: it explains chromosomes as carriers of genetic information, meiosis as the mechanism that halves chromosome number into haploid gametes, and the random independent assortment of 23 chromosome pairs as the source of the enormous variation among offspring. Activity 11.4 directly operationalises chromosome segregation by simulating gamete formation with bead combinations. The section explains how this variation enables adaptation and evolution — meeting C-3.3's 'variation' clause. This strand is independent from the reproductive-process mechanics of C-3.2 and carries co-central weight. |

## Ch 12 Patterns in Life: Diversity and Classification

15×50min · 15 periods · 750min

| # | Code         | Competency                                                                                                                                                                       | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-4.1</b> | Applies the knowledge of cellular diversity in organisms along with the ecological role organisms play (autotrophic/heterotrophic nutrition) to classify them into five-kingdoms | The chapter's dominant strand delivers the five-kingdom classification using cellular diversity and ecological roles as the organising criteria — cell type (prokaryote/eukaryote), level of organisation (unicellular/multicellular), cell wall composition and mode of nutrition (autotrophic/heterotrophic/decomposer) — applied systematically across Monera, Protista, Fungi, Plantae (five classes) and Animalia (eight invertebrate phyla, protochordates and five vertebrate groups). This is precisely the scope of C-4.1: applying knowledge of cellular diversity and ecological role to classify organisms into the five kingdoms. |

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|---|--------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | <b>C-4.2</b> | Illustrates different levels of organisations of living organisms (from molecules to organisms) | A distinct, sectioned second strand develops levels of biological organisation. The invertebrate sequence (sections 12.6.5 Porifera through Echinodermata) is explicitly structured around the progression from cellular-level organisation (Porifera) → tissue level (Cnidaria) → organ level (Platyhelminthes) → organ system level (Nematoda through Echinodermata), and section 12.7.1 introduces the hierarchical classification pyramid from Kingdom down to Species. Together these constitute an independent, cognitively distinct treatment of C-4.2 — illustrating levels of organisation from molecules to organisms — that runs alongside but is separable from the five-kingdom classification strand. |
|---|--------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Ch 13 Earth as a System: Energy, Matter, and Life

9×50min · 9 periods · 450min

| # | Code         | Competency                                                                                                                                         | Justification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>C-4.3</b> | Analyses different levels of biological organisation from organisms to ecosystems and biomes along with interactions that take place at each level | The chapter's entire domain is the analysis of how living organisms interact with their abiotic environment at the ecosystem level. It frames the Earth as five interacting spheres (geosphere, hydrosphere, cryosphere, atmosphere, biosphere) and traces how energy flow (solar radiation → wind → ocean currents) and matter cycling (water, carbon, nitrogen, oxygen biogeochemical cycles) interconnect biotic and abiotic components. Section 13.3 on biogeochemical cycles is an explicit treatment of the transfer of matter and energy between abiotic and biotic components that sustains ecosystems; Section 13.4 analyses how human disruption of these cycles cascades across all spheres. No other CG-4 competency is addressed: C-4.1 (five-kingdom classification), C-4.2 (molecular-to-organism levels of organisation), C-4.4 (Mendelian inheritance), and C-4.5 (natural selection and evolution) are all absent. |